

# Integrative Structure Validation Report ?

January 07, 2026 - 10:34 AM PST

*The following software was used in the production of this report:*

*IHMValidation* Version 3.0

*Python-IHM* Version 2.7

*MolProbity* Version 4.5.2

|                   |                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------|
| PDB ID            | 8ZZU   pdb_00008zzu                                                                            |
| PDB-Dev ID        | PDBDEV_00000030                                                                                |
| Structure Title   | Molecular Architecture of Human Fibrin Clots                                                   |
| Structure Authors | Oleg Klykov; Carmen van der Zwaan; Albert J.R. Heck; Alexander B. Meijer; Richard A. Scheltema |
| Deposited on      | 2019-03-14                                                                                     |

*This is a PDB-IHM Structure Validation Report.*

We welcome your comments at [helpdesk@pdb-ihm.org](mailto:helpdesk@pdb-ihm.org)

A user guide is available at [https://pdb-ihm.org/validation\\_help.html](https://pdb-ihm.org/validation_help.html) with specific help available everywhere you see the ? symbol.

List of references used to build this report is available [here](#).

## 1. Overview ?

### 1.1. Summary ?

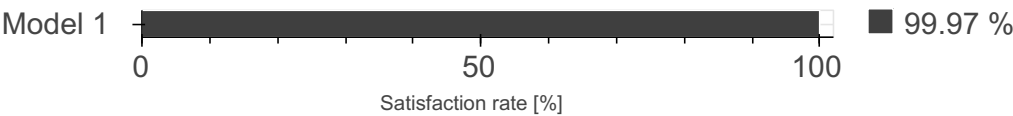
*This entry consists of 1 model(s). A total of 9 dataset(s) were used to build this entry.*

| Name                 | Type              | Count |
|----------------------|-------------------|-------|
| Crosslinking-MS data | Experimental data | 1     |
| Experimental model   | Starting model    | 1     |
| De Novo model        | Starting model    | 7     |

1.2. Overall quality ?

This validation report contains model quality assessments for all structures, data quality and fit to model assessments for SAS and crosslinking-MS datasets. Data quality and fit to model assessments for other datasets and model uncertainty are under development. Number of plots is limited to 256.

Model Quality: Excluded Volume Analysis ?



Fit to Data Used for Modeling ?

Crosslink satisfaction



2. Model Details ?

2.1. Ensemble information ?

This entry consists of 0 distinct ensemble(s).

2.2. Representation ?

This entry has 1 representation(s).

| ID | Model(s) | Entity ID | Molecule name                             | Chain(s) [auth] | Total residues | Rigid segments | Flexible segments | Model coverage/<br>Starting model coverage (%) | Scale  |
|----|----------|-----------|-------------------------------------------|-----------------|----------------|----------------|-------------------|------------------------------------------------|--------|
| 1  | 1        | 1         | RGD-containing Alpha Domain Alpha_558-620 | A [I]           | 63             | 13-35, 53-63   | 1-12, 36-52       | 100.00 / 100.00                                | Atomic |
|    |          |           |                                           | M [J]           |                |                |                   |                                                |        |
|    |          | 2         | Subunit Alpha_46-219                      | B [A]           | 174            | 1-174          | -                 | 100.00 / 100.00                                | Atomic |
|    |          |           |                                           | G [D]           |                |                |                   |                                                |        |
|    |          | 3         | Subunit Alpha_220-249                     | C [Z]           | 30             | 1-30           | -                 | 100.00 / 100.00                                | Atomic |
|    |          |           |                                           | H [Y]           |                |                |                   |                                                |        |
|    |          | 4         | Subunit Beta                              | D [B]           | 401            | 1-401          | -                 | 100.00 / 100.00                                | Atomic |
|    |          |           |                                           | J [E]           |                |                |                   |                                                |        |
|    |          | 5         | Subunit Gamma                             | E [C]           | 381            | 1-381          | -                 | 100.00 / 100.00                                | Atomic |
|    |          |           |                                           | K [F]           |                |                |                   |                                                |        |
|    |          | 6         | Interactive Alpha Domain Alpha_432-491    | F [H]           | 60             | 29-60          | 1-28              | 100.00 / 100.00                                | Atomic |
|    |          |           |                                           |                 |                |                |                   |                                                |        |

| ID | Model(s) | Entity ID | Molecule name          | Chain(s) [auth] | Total residues | Rigid segments | Flexible segments | Model coverage/<br>Starting model coverage (%) | Scale  |
|----|----------|-----------|------------------------|-----------------|----------------|----------------|-------------------|------------------------------------------------|--------|
|    |          |           |                        | L [G]           |                |                |                   |                                                |        |
|    |          | 7         | Beta N-term Beta_55-84 | I [K]           | 34             | -              | 1-34              | 100.00 / 100.00                                | Atomic |

### 2.3. Datasets used for modeling ?

There are 9 unique datasets used to build the models in this entry.

| ID | Dataset type         | Database name | Data access code             |
|----|----------------------|---------------|------------------------------|
| 1  | Experimental model   | PDB           | <a href="#">pdb_00003ghg</a> |
| 2  | De Novo model        | Not available | Not available                |
| 3  | De Novo model        | Not available | Not available                |
| 4  | De Novo model        | Not available | Not available                |
| 5  | De Novo model        | Not available | Not available                |
| 6  | Crosslinking-MS data | PRIDE         | <a href="#">PXD011680</a>    |
| 7  | De Novo model        | Not available | Not available                |
| 8  | De Novo model        | Not available | Not available                |
| 9  | De Novo model        | Not available | Not available                |

### 2.4. Methodology and software ?

This entry is a result of 1 distinct protocol(s).

| Step number | Protocol ID | Method name   | Method type   | Method description | Number of computed models | Multi state modeling | Multi scale modeling |
|-------------|-------------|---------------|---------------|--------------------|---------------------------|----------------------|----------------------|
| 1           | 1           | Not available | Not available | Not available      | Not available             | False                | False                |

There are 8 software packages reported in this entry.

| ID | Software name                | Software version | Software classification | Software location                                                                                                                               |
|----|------------------------------|------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <a href="#">ThreaDomEx</a>   | Not available    | Not available           | <a href="https://zhanglab.ccmb.med.umich.edu/ThreaDomEx/">https://zhanglab.ccmb.med.umich.edu/ThreaDomEx/</a>                                   |
| 2  | <a href="#">Robetta</a>      | Not available    | model building          | <a href="http://robetta.bakerlab.org">http://robetta.bakerlab.org</a>                                                                           |
| 3  | <a href="#">HAlign-Kbest</a> | Not available    | model building          | <a href="http://bioserv.rpbs.univ-paris-diderot.fr/services/HAlign-Kbest/">http://bioserv.rpbs.univ-paris-diderot.fr/services/HAlign-Kbest/</a> |
| 4  | <a href="#">I-TASSER</a>     | Not available    | model building          | <a href="https://zhanglab.ccmb.med.umich.edu/I-TASSER/">https://zhanglab.ccmb.med.umich.edu/I-TASSER/</a>                                       |
| 5  | <a href="#">RaptorX</a>      | Not available    | model building          | <a href="http://raptorx.uchicago.edu">http://raptorx.uchicago.edu</a>                                                                           |
| 6  | <a href="#">HADDOCK</a>      | 2.20             | model building          | <a href="https://haddock.science.uu.nl">https://haddock.science.uu.nl</a>                                                                       |
| 7  | <a href="#">CPORT</a>        | Not available    | Not available           | <a href="https://milou.science.uu.nl/services/CPORT/">https://milou.science.uu.nl/services/CPORT/</a>                                           |

| ID | Software name          | Software version | Software classification | Software location                                                                                                             |
|----|------------------------|------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 8  | <a href="#">DisVis</a> | Not available    | Not available           | <a href="https://milou.science.uu.nl/cgi/services/DISVIS/disvis/">https://milou.science.uu.nl/cgi/services/DISVIS/disvis/</a> |

### 3. Data quality ?

#### 3.2. Crosslinking-MS

At the moment, data validation is only available for crosslinking-MS data deposited as a fully [compliant](#) dataset in the [PRIDE Crosslinking](#) database. Correspondence between crosslinking-MS and entry entities is established using [pyHMMER](#). Only residue pairs that passed the reported threshold are used for the analysis. The values in the report have to be interpreted in the context of the experiment (i.e. only a minor fraction of in-situ or in-vivo dataset can be used for modeling).

Crosslinking-MS dataset is not available in the [PRIDE Crosslinking](#) database.

### 4. Model quality ?

For models with atomic structures, MolProbity analysis is performed. For models with coarse-grained or multi-scale structures, excluded volume analysis is performed.

#### 4.1a. Excluded Volume Analysis ?

Excluded volume satisfaction for the models in the entry are listed below. The Analysed column shows the number of particle-particle or particle-atom pairs for which excluded volume was analysed.

| Model ID | Analysed  | Number of violations | Excluded Volume Satisfaction (%) |
|----------|-----------|----------------------|----------------------------------|
| 1        | 247720411 | 85135                | 99.97                            |

### 5. Fit to Data Used for Modeling Assessment ?

#### 5.2. Crosslinking-MS ?

##### 5.2.1. Restraint types ?

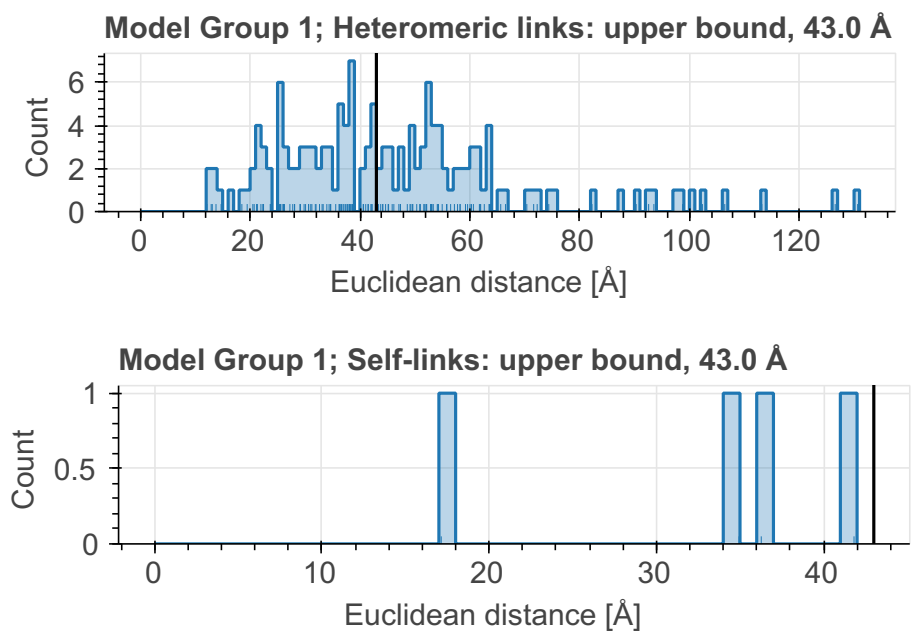
This table summarizes information about crosslinker(s) used for data generation, and how crosslinking information was translated into actual modeling restraints. Restraints assigned "[by-residue](#)" are interpreted as between CA atoms. Restraints between coarse-grained beads are indicated as "[coarse-grained](#)". [Restraint group](#) represents a set of crosslinking restraints applied collectively in the modeling.

There are 156 crosslinking restraints combined in 156 restraint groups.

| Linker | Residue 1 | Atom 1 | Residue 2 | Atom 2 | Restraint type | Distance, Å | Count |
|--------|-----------|--------|-----------|--------|----------------|-------------|-------|
| DSSO   | LYS       | CA     | LYS       | CA     | upper bound    | 43.00       | 156   |

[Distograms of individual restraints](#)

Distograms (i.e., histogram plots of distances) provide an overview of distributions of distances between residues for which chemical crosslinks were identified. The shift of the distogram relative to the threshold value may indicate a poor model. Restraints with identical thresholds are grouped into one plot. Only the best distance per restraint per model group/ensemble is plotted. Inter- and intramolecular (including self-links) restraints are also grouped into one plot. Distance for a restraint between coarse-grained beads is calculated as a minimal distance between shells; if beads intersect, the distance will be reported as 0.0. A bead with the highest available resolution for a given residue is used for the assessment.



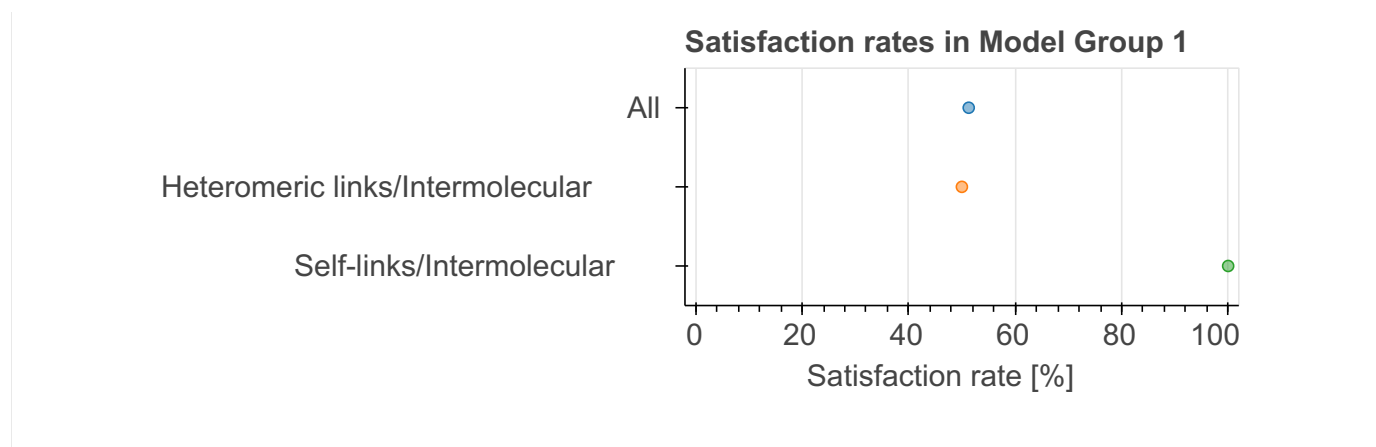
5.2.2. Satisfaction of restraints ?

Satisfaction of restraints is calculated on a [restraint group](#) (a set of crosslinking restraints applied collectively in the modeling) level. Satisfaction of a restraint group depends on satisfaction of individual restraints in the group and the conditionality (all/any). A restraint group is considered satisfied, if the condition was met in at least one model of the model group/ensemble. The number of measured restraints can be smaller than the total number of restraint groups if crosslinks involve non-modeled residues. Only deposited models are used for validation right now.

| State group | State | Model group | # of Deposited models/Total | Restraint group type              | Satisfied (%) | Violated (%) | Count (Total=156) |
|-------------|-------|-------------|-----------------------------|-----------------------------------|---------------|--------------|-------------------|
| 1           | 1     | 1           | 1/1                         | All                               | 51.28         | 48.72        | 156               |
|             |       |             |                             | Heteromeric links/ Intermolecular | 50.00         | 50.00        | 152               |
|             |       |             |                             | Self-links/ Intermolecular        | 100.00        | 0.00         | 4                 |

Per-model satisfaction rates in ensembles

Every point represents one model in a model group/ensemble. Where possible, boxplots with quartile marks are also plotted.



## 6. Fit to Data Used for Validation Assessment ?

Validation for this section is under development.

### Acknowledgments

The development of integrative model validation metrics, implementation of a model validation pipeline, and creation of a validation report for integrative structures are funded by NSF awards to the [PDB-IHM team](#) (DBI-1756248, DBI-2112966, DBI-2112967, DBI-2112968, and DBI-1756250) and awards from NSF, NIH, and DOE to the [RCSB PDB](#) (DBI-2321666, R01GM157729, and DE-SC0019749). The PDB-IHM team and members of the [Sali lab](#) contributed model validation metrics and software packages.

Dr. Jill Trehwella, Dr. Dina Schneidman, and members of the [SASBDB](#) repository are acknowledged for their advice and support in implementing SAS validation methods. Team members from the labs of Dr. Juri Rappsilber, Dr. Alexander Leitner, Dr. Andrea Graziadei, and members of [PRIDE](#) database are acknowledged for their advice and support in implementing crosslinking-MS validation methods. We are grateful to Dr. Shruthi Viswanath for discussions about uncertainty assessment of integrative structural models.

Members of the [wwPDB Integrative/Hybrid Methods Task Force](#) provided recommendations and community support for the project.